

Geometric Quantities for the Schwarzschild Metric

Generated by EinsteinKPy2Tex

June 6, 2026

1 Citation Information

If you use these results and code in your research, please cite:

- Einstein, A. (1915). Die Feldgleichungen der Gravitation. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, 844-847.
- Ribeiro, S., Bapat, A., Tandon, A., & EinsteinPy Developers (2020). EinsteinPy: A Python Package for General Relativity. *Journal of Open Source Software*, 5(51), 2356. <https://doi.org/10.21105/joss.02356>
- P. Ntelis (2026). EinsteinKPy2Tex: Schwarzschild Metric Tensor Calculator (Version 1.0). <https://github.com/lontelis/EinsteinKPy2Tex>

2 Geometric quantities for the Schwarzschild metric

The Schwarzschild metric is given by

$$ds^2 = - \left(1 - \frac{2GM}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2. \quad (1)$$

2.1 Metric tensor $g_{\mu\nu}$

$$g_{\mu\nu} = \begin{bmatrix} -c^2 \left(-\frac{R_s}{r} + 1\right) & 0 & 0 & 0 \\ 0 & \frac{1}{-\frac{R_s}{r} + 1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} \quad (2)$$

2.2 Christoffel symbols $\Gamma_{\nu\rho}^{\mu}$ (non-zero)

$$\Gamma_{tr}^t = \frac{R_s}{2r(-R_s + r)} \quad (3)$$

$$\Gamma_{rt}^t = \frac{R_s}{2r(-R_s + r)} \quad (4)$$

$$\Gamma_{tt}^r = \frac{R_s c^2 (-R_s + r)}{2r^3} \quad (5)$$

$$\Gamma_{rr}^r = \frac{R_s}{2r(R_s - r)} \quad (6)$$

$$\Gamma_{\theta\theta}^r = R_s - r \quad (7)$$

$$\Gamma_{\phi\phi}^r = (R_s - r) \sin^2(\theta) \quad (8)$$

$$\Gamma_{r\theta}^{\theta} = \frac{1}{r} \quad (9)$$

$$\Gamma_{\theta r}^{\theta} = \frac{1}{r} \quad (10)$$

$$\Gamma_{\phi\phi}^{\theta} = -\frac{\sin(2\theta)}{2} \quad (11)$$

$$\Gamma_{r\phi}^{\phi} = \frac{1}{r} \quad (12)$$

$$\Gamma_{\theta\phi}^{\phi} = \frac{1}{\tan(\theta)} \quad (13)$$

$$\Gamma_{\phi r}^{\phi} = \frac{1}{r} \quad (14)$$

$$\Gamma_{\phi\theta}^{\phi} = \frac{1}{\tan(\theta)} \quad (15)$$

$$(16)$$

2.3 Riemann curvature tensor $R^\mu_{\nu\rho\sigma}$ (non-zero)

$$R^t_{rtr} = \frac{R_s}{r^2 (-R_s + r)} \quad (17)$$

$$R^t_{rrt} = \frac{R_s}{r^2 (R_s - r)} \quad (18)$$

$$R^t_{\theta t \theta} = -\frac{R_s}{2r} \quad (19)$$

$$R^t_{\theta \theta t} = \frac{R_s}{2r} \quad (20)$$

$$R^t_{\phi t \phi} = -\frac{R_s \sin^2(\theta)}{2r} \quad (21)$$

$$R^t_{\phi \phi t} = \frac{R_s \sin^2(\theta)}{2r} \quad (22)$$

$$R^r_{ttr} = \frac{R_s c^2 (-R_s + r)}{r^4} \quad (23)$$

$$R^r_{trt} = \frac{R_s c^2 (R_s - r)}{r^4} \quad (24)$$

$$R^r_{\theta r \theta} = -\frac{R_s}{2r} \quad (25)$$

$$R^r_{\theta \theta r} = \frac{R_s}{2r} \quad (26)$$

$$R^r_{\phi r \phi} = -\frac{R_s \sin^2(\theta)}{2r} \quad (27)$$

$$R^r_{\phi \phi r} = \frac{R_s \sin^2(\theta)}{2r} \quad (28)$$

$$R^\theta_{tt\theta} = \frac{R_s c^2 (R_s - r)}{2r^4} \quad (29)$$

$$R^\theta_{t\theta t} = \frac{R_s c^2 (-R_s + r)}{2r^4} \quad (30)$$

$$R^\theta_{rr\theta} = \frac{R_s}{2r^2 (-R_s + r)} \quad (31)$$

$$R^\theta_{r\theta r} = \frac{R_s}{2r^2 (R_s - r)} \quad (32)$$

$$R^\theta_{\phi \theta \phi} = \frac{R_s \sin^2(\theta)}{r} \quad (33)$$

$$R^\theta_{\phi \phi \theta} = -\frac{R_s \sin^2(\theta)}{r} \quad (34)$$

$$R^\phi_{tt\phi} = \frac{R_s c^2 (R_s - r)}{2r^4} \quad (35)$$

$$R^\phi_{t\phi t} = \frac{R_s c^2 (-R_s + r)}{2r^4} \quad (36)$$

$$R^\phi_{rr\phi} = \frac{R_s}{2r^2 (-R_s + r)} \quad (37)$$

$$R^\phi_{r\phi r} = \frac{3 R_s}{2r^2 (R_s - r)} \quad (38)$$

$$R^\phi_{\theta \theta \phi} = -\frac{R_s}{r} \quad (39)$$

$$R^\phi_{\theta \phi \theta} = \frac{R_s}{r} \quad (40)$$

$$(41)$$

2.4 Ricci tensor $R_{\mu\nu}$ (non-zero)

$$\text{All components zero.} \quad (42)$$

2.5 Ricci scalar R

$$0 \quad (43)$$

2.6 Einstein tensor $G_{\mu\nu}$ (non-zero)

$$\text{All components zero.} \quad (44)$$

2.7 Geodesic equations

The geodesic equations are given by:

$$\ddot{t} = -\frac{R_s \dot{r} \dot{t}}{r(-R_s + r)} \quad (45)$$

$$\ddot{r} = -\frac{R_s \dot{r}^2}{2r(R_s - r)} - \frac{R_s \dot{t}^2 c^2 (-R_s + r)}{2r^3} - \dot{\phi}^2 (R_s - r) \sin^2(\theta) - \dot{\theta}^2 (R_s - r) \quad (46)$$

$$\ddot{\theta} = \frac{\dot{\phi}^2 \sin(2\theta)}{2} - \frac{2\dot{\theta} \dot{r}}{r} \quad (47)$$

$$\ddot{\phi} = -\frac{2\dot{\phi} \dot{\theta}}{\tan(\theta)} - \frac{2\dot{\phi} \dot{r}}{r} \quad (48)$$

$$(49)$$

where λ is an affine parameter and $\dot{x}^\mu = dx^\mu/d\lambda$.

2.8 Kretschmann scalar

The Kretschmann scalar $K = R^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}$ is:

Full expression (unsimplified)

$$\frac{R_s^2 c^4 (-R_s + r)^2 \left(-\frac{R_s}{r} + 1\right)^2}{r^4 (R_s - r)^2 (-R_s c^2 + c^2 r)^2} + \frac{2R_s^2 c^4 \left(-\frac{R_s}{r} + 1\right)^2}{r^4 (-R_s c^2 + c^2 r)^2} + \frac{R_s^2 c^4 (-R_s + r)^2}{2r^6 (-R_s c^2 + c^2 r)^2} + \frac{R_s^2 c^4 (R_s - r)^2}{2r^6 (-R_s c^2 + c^2 r)^2} + \frac{R_s^2 c^4}{r^8 (-R_s c^2 + c^2 r)^2} \quad (50)$$

2.8.1 Simplified expression from code

$$K = \frac{12R_s^2}{r^6} \tag{51}$$

Note: The simplified expression should equal the analytic result from the literature, i.e. $K = \frac{48G^2M^2}{c^4r^6} = \frac{12R_s^2}{r^6}$. If they do not match, check the Riemann tensor index ordering.